DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY

UTTAR PRADESH , LUCKNOW



Evaluation Scheme & Syllabus

MBA (Business Analytics)

First Year

AS PER

AICTE MODEL CURRICULUM

& NEP 2020

(Effective from the Session: 2024-25)

MBA (Business Analytics) 1st Year Course Structure in accordance with AICTE Model Curriculum & NEP 2020 Effective w.e.f. Academic Session 2024-25

SEMESTER I

SN	CODE	SUBJECT	PERIODS			INTERNAL EVALUATION SCHEME				END SEMESTER		TOTAL	CREDIT
			L	T	P	CT	TA	PS	TOTAL	TE	PE		
1	BMBA101	Management Concepts & Organizational Behaviour	4	0	0	20	10	0	30	70	0	100	3
2	BMBA102	Managerial Economics	4	0	0	20	10	0	30	70	0	100	3
3	BMBA103	Financial Accounting and Analysis	3	1	0	20	10	0	30	70	0	100	3
4	BMBA104	Business Statistics and Analytics	3	1	0	20	10	0	30	70	0	100	3
5	BMBA105	Marketing Management	4	0	0	20	10	0	30	70	0	100	3
6	BMBA106	Introduction to Business Analytics & Data Science	4	0	0	20	10	0	30	70	0	100	3
7	BMBA151	IT Skills-1	0	0	4	0	0	30	30	0	70	100	3
8	BMBA152	Basics of Data Management With “R”	1	0	3	0	0	30	30	0	70	100	3
9	BMBA153	Business Intelligence Using Power BI	0	0	3	0	0	30	30	0	70	100	2
10	BVA 151	SPORTS & YOGA *	0	0	3	0	100	0	100	0	0	100	0
												1000	26

***Qualifying but Non Credit Course**

SEMESTER II

SN	CODE	SUBJECT	PERIODS			INTERNAL EVALUATION SCHEME				END SEMESTER		TOTAL	CREDIT
			L	T	P	CT	TA	PS	TOTAL	TE	PE		
1	BMBA201	Exploratory and Predictive Analytics Using Python	2	0	2	20	10	0	30	70	0	100	3
2	BMBA202	Data Mining Techniques – Predictive Modeling& Pattern Discovery- Using R	2	0	2	20	10	0	30	70	0	100	3
3	BMBA203	Business Research Methods	4	0	0	20	10	0	30	70	0	100	3
4	BMBA204	Financial Management & Corporate Finance	4	0	0	20	10	0	30	70	0	100	3
5	BMBA205	Database Management System and SQL	2	0	2	20	10	0	30	70	0	100	3
6	BMBA206	Quantitative Techniques for Managers	3	1	0	20	10	0	30	70	0	100	3
7	BMBA251	Data Visualization Using Tableau	0	0	4	0	0	30	30	0	70	100	3
8	BMBA252	Analytics Project using R / Python	0	0	3	0	0	30	30	0	70	100	2
9	BMBA253	IT Skills-2	0	0	2	0	0	30	30	0	70	100	1
10	BMBA254	MINI PROJECT	0	0	3	0	0	30	30	0	70	100	2
												1000	26

L/T/P – Lecture/Tutorial/Practical, CT/TA/PS- Class Test/Teachers Assessment/Practical Session, TE/PE- Term End/ Practical End

SEMESTER I

MANAGEMENT CONCEPTS AND ORGANISATIONAL BEHAVIOUR

BMBA101

Course Credit: 3 **Contact Hours: 40**

Course Objectives:

1. To provide fundamentals of management concepts & theories
2. To help the students understand the concepts of organisational behaviour
3. To apply the concepts of management and organisational behaviours in real-world situations
4. Familiarize the students with the contemporary challenges and prospects in management.
5. Developing managerial and leadership skills among students

UNIT I (8 Lectures)

Fundamentals of Management: Management practices from past to present, Different levels of management, Managerial skills and Managerial Functions, Case Studies

Planning- Objective of planning, Planning process, Types of planning, Types of plans, Management by Objective, Decision-making- types, process & techniques, Case Studies

UNIT-II (8 Lectures)

Organizing & Staffing- Types of organisation, Organization structure, delegation and decentralisation of authority, Meaning of staffing, Recruitment, selection & placement, Training & development.

Directing & Controlling- Principle of directing, Essence of coordination, Different control techniques, Management by exception. Case Studies

UNIT III (8 Lectures)

Fundamentals of individual behavior- Concepts of OB, OB Model, Components of OB, Personality, types of personality, Personal effectiveness, meaning of Attitudes, Types, Components, attitude formation and attitude change. Meaning & Type of Group Behaviour, Interpersonal skills, Transactional Analysis, Johari Window, Meaning of Perception, process, behavioral applications of perception. Case Studies.

UNIT IV (8 Lectures)

Motivation: Theory of Motivation: Maslow's, Herzberg's, McClelland, Contemporary theories of Motivation: Self Determination Theory, Self-Efficacy Theory, Vroom's Expectancy Theory, Equity Theory, Reinforcement Theory.

UNIT V: (8 Lectures)

Leadership: What is leadership, types of leaders and leadership styles, traits and qualities of effective leader, trait theory, LSM – Leadership Situational Model, Team Building, Tuckman Model of Team Development. Organizational Change: Meaning of organizational change approaches to managing organizational change, creating a culture for change, implementing the change, Kurt Lewin Model of change. Case Studies

COURSE OUTCOME

Course Outcomes	Bloom's taxonomy
CO 1: Developing understanding of managerial practices and their perspectives.	Knowledge (K 2) Remembering (K1)
CO2: Understanding and applying the concepts of Organizational Behaviour	Knowledge (K 2) Applying (K 4)
CO 3: Applying the concepts of management and analyze organizational behaviors in real world situations	Applying (K 4) Analyzing (K 5)
CO 4: Comprehend and practice contemporary issues in management.	Comprehending (K 3)
CO 5: Applying managerial and leadership skills among students	Applying (K4)

Suggested Readings

1. Koontz Harold &Weihrich Heinz – Essentials of management (Tata McGraw Hill, 5th Edition,2008)
2. L. M. Prasad- Principles and Practices of Management, Sulatn Chand & Sons, 7th edition, 2007.
3. Stephen P. Robbins, ―Organizational Behaviour‖, 12th Edition, Prentice Hall
4. Dr. Premvir Kapoor, Principles and Practices of Management, Khanna Publishing House, Delhi
5. Robbins & Coulter - Management (Prentice Hall of India, 9th Edition)
6. Principles of Management, George R. Terry & S.G. Franklin, AITBS, Delhi.
7. N M Khandelwal- Indian Ethos & Values for Management- Himalyan Publishing
8. Fred Luthans, ―Organizational Behaviour‖, 12th Edition, McGraw Hill International Edition
9. Aswathappa K, ―Organizational Behaviour (Text, Cases and Games) ‖, Himalaya Publication
10. UdaiPareek, ―Organizational Behavior‖, Oxford University Press

MANAGERIAL ECONOMICS
BMBA102

Course Credit: 3 Contact Hours: 40

Course Objective:

1. To understand the importance of Managerial Economics in management and businesses
2. To apply the principles of managerial economics in achieving business objectives
3. Be equipped with the tools necessary in forecasting product demand
4. Understand and be able to apply latest pricing strategies
5. Understand and analyze the macro environment affecting the business decision making.

UNIT –I (6 Hours)

Basic Concepts and principles: Definition, Nature and Scope of Economics-Micro Economics and Macro Economics, Managerial Economics and its relevance in business decisions. Fundamental Principles of Managerial Economics - Incremental Principle, Marginal Principle, Opportunity Cost Principle, Discounting Principle, Concept of Time Perspective, Equi-Marginal Principle, Utility Analysis, Cardinal Utility and Ordinal Utility. Case Studies

UNIT –II (8Hours)

Demand and Supply Analysis: Theory of Demand, Types of Demand. Determinants of demand, Demand Function, Demand Schedule, Demand curve, Law of Demand, Exceptions to the law of Demand, Shifts in demand curve, Elasticity of Demand and its measurement. Price Elasticity, Income Elasticity, Arc Elasticity. Cross Elasticity and Advertising Elasticity. Uses of Elasticity of Demand for managerial decision making, Demand forecasting meaning, significance and methods. (Numerical Exercises) Case Studies
Supply Analysis; Law of Supply, Supply Elasticity; Analysis and its uses for managerial decision making, Price of a Product under demand and supply forces. Case Studies

UNIT –III (10Hours)

Production and cost Analysis: Production concepts & analysis; Production function, Types of production function, Laws of production: Law of diminishing returns, Law of returns to scale.
Cost concept and analysis: Cost, Types of costs, Cost output relationship in the short-run. Cost output relationship in the Long-run. Estimation of revenue. Average Revenue, Marginal Revenue. Case Studies

UNIT –IV (10Hours)

Market structures: Perfect and Imperfect Market Structures, Perfect Competition, features, determination of price under perfect competition. Monopoly: Feature, pricing under monopoly, Price Discrimination. Monopolistic: Features, pricing under monopolistic competition, product differentiation. Oligopoly: Features, kinked demand curve, cartels, price leadership. Case Studies

UNIT –V (6Hrs)

National Income; Concepts and various methods of its measurement, Circular flows in 2 sector, 3 sector, 4 sector economies, Inflation, types and causes, Business Cycle & its phases.

Course Outcomes:

Course Outcomes	Bloom's taxonomy
CO1: Students will be able to remember the concepts of micro economics and also able to understand the various micro economic principles to make effective economic decisions under conditions of risk and uncertainty.	• Knowledge (K 2) • Remembering (k1)
CO2: The students would be able to understand the law of demand & supply & their elasticities, evaluate & analyse these concepts and apply them in various changing situations in industry. Students would be able to apply various techniques to forecast demand for better utilization of resources.	• Knowledge (K 2) • Applying (K 4) • Synthesizing (K6) • Evaluating (K7)
CO3: The students would be able to understand the production concept and how the production output changes with the change in inputs and able to analyse the effect of cost to business and their relation to analyze the volatility in the business world	• Comprehending (K 3) • Applying (K 4) • Analyzing (K 5) • Evaluating (K7)
CO4: The students would be able to understand & evaluate the different market structure and their different equilibriums for industry as well as for consumers for the survival in the industry by the application of various pricing strategic	• Applying (K 4) • Analyzing (K 5) • Synthesizing (K6)
CO5: The students would be able to analyse the macroeconomic concepts & their relation to micro economic concept & how they affect the business & economy.	• Knowledge (K 2) • Comprehending (K 3)

Suggested Readings

1. Managerial Economics, D. N. Dwivedi,Vikas Publication, 7th Ed
2. Managerial Economics, GEETIKA, McGraw-Hill Education 2nd Ed.
3. Managerial Economics: Concepts and Applications (SIE), THOMAS& MAURICE, McGraw-Hill Education, 9th Ed
4. Managerial Economics, H. L Ahuja, S.Chand, 8th Ed
5. Managerial Economics – Theory and Applications, Dr. D.M. Mithani, Himalaya Publications, 7th Ed.
6. Sociology & Economics for Engineers, Dr. Premvir Kapoor, Khanna Publishing House

FINANCIAL ACCOUNTING AND ANALYSIS
BMBA103

Course Credit: 3 **Contact Hours: 40**

Course Objectives:

1) To understand the fundamentals, basic theory and concepts of financial accounting.
2) To have a knowledge about various Accounting Standards used in preparation of financial statements.
3) To have an understanding of preparation and presentation of financial statements.
4) To acquire knowledge about various techniques used for analysing financial statements with its application.
5) To enable students acquainted with current trends and social responsibility accounting.

UNIT I (4 Hrs)

Meaning and Scope of Accounting: Evolution and Users of Accounting, Basic Accounting terminologies, Principles of Accounting, Accounting Concepts & Conventions, Accounting Equation, Deprecation Accounting. GAAP (Introduction).

UNIT II (10 Hrs)

Mechanics of Accounting: Accounting Standards and IFRS: International Accounting Principles and Standards; Matching of Indian Accounting Standards with International Accounting Standards, Double entry system of Accounting, journalizing of transactions; Ledger posting and Trial Balance.

UNIT III (12 Hrs)

Presentation of Financial Statement: Preparation of final accounts (Profit & Loss Account and Balance Sheet) according to companies act 2013 (vertical format), Excel Application to make Balance sheet, Case studies and Workshops, Preparation of Cash Flow Statement and its analysis.

UNIT IV (8 Hrs)

Analysis of financial statement: Ratio Analysis- Solvency ratios, Profitability ratios, activity ratios, liquidity ratios, Market capitalization ratios; leverage Ratio, Detailed Analysis using excel application.

UNIT V (6 Hrs)

Financial Statement Analysis and Recent Types of Accounting: Common Size Statement; Comparative Balance Sheet and Trend Analysis of manufacturing, Service & banking organizations, Case Study and Workshops in analysing Balance sheet. Human Resource Accounting, Forensic Accounting, Accounting for corporate social responsibility.

Course Outcome:

After successful completion of this course students will be able to

S.No.	Course Outcome	Bloom's Taxonomy
1	CO1.Understand and apply accounting concepts, principles and conventions for their routine monetary transaction.	Knowledge (K2)/ Comprehending (K 3)
2	CO2. Understand about IFRS, Ind AS and IAS for preparation and reporting of financial statements.	Knowledge (K2) Synthesizing (K6)/
3	CO3. Create and prepare financial statements and Cash flow in accordance with Generally Accepted Accounting Principles	Remembering (k1)
4	CO4. Analyse, interpret and communicate the information contained in basic financial statements and explain the limitations of such statements.	Analyzing (K 4) / Evaluating (K7))
5	CO5. Recognizing various types of accounting and utilize the technology and social responsibility in facilitating and enhancing accounting and financial reporting processes	Knowledge (K2) Applying (K 4)

Suggested Readings

1. Maheshwari S.N &Maheshwari S K – A text book of Accounting for Management (Vikas, 10th Edition)
2. Essentials of Financial Accounting (based on IFRS), Bhattacharya (PHI,3rd Ed)
3. Khan and Jain - Financial Management (Tata McGraw Hill, 7th Ed.)
4. PC Tulsian- Financial Accounting (Pearson, 2016)
5. Dhamija - Financial Accounting for managers: (Prentice Hall, 2nd Edition).
6. Narayanswami - Financial Accounting: A Managerial Perspective (PHI,5th Ed)
7. Dhanesh K Khatri- Financial Accounting (TMH,2015)
8. Ambrish Gupta - Financial Accounting: A Managerial Perspective (Prentice Hall, 4th Edition)
9. Ramchandran & Kakani - Financial Accounting for Management (TMH, 2nd Edition).
10. Mukherjee - Financial Accounting for Management (TMH, 2nd Edition).

BUSINESS STATISTICS AND ANALYTICS
BMBA104

Course Credit: 3 **Contact Hours: 40 hours**

Course Objectives

1. Understand the different basic concept / fundamentals of business statistics.
2. Understand the importance of measures of Descriptive statistics which includes measures of central tendency, Measures of Dispersion, Time Series Analysis, Index Number, Correlation and Regression analysis and their implication on Business performance.
3. Understand the concept of Probability and its usage in various business applications.
4. Understand the Hypothesis Testing concepts and use inferential statistics- t, F, Z Test and Chi Square Test
5. Understand the practical application of Descriptive and Inferential Statistics concepts and their uses for Business Analytics.

Unit I (10 Sessions): Descriptive Statistics
Meaning, Scope, types, functions and limitations of statistics, Measures of Central tendency – Mean, Median, Mode, Quartiles, Measures of Dispersion – Range, Inter quartile range, Mean deviation, Standard deviation, Variance, Coefficient of Variation, Skewness and Kurtosis.

Unit II (8 Sessions): Time Series & Index Number
Time series analysis: Concept, Additive and Multiplicative models, Components of time series, Trend analysis: Least Square method - Linear and Non- Linear equations, Applications in business decision-making.
Index Numbers: - Meaning, Types of index numbers, uses of index numbers, Construction of Price, Quantity and Volume indices: - Fixed base and Chain base methods.

Unit III (6 Sessions): Correlation & Regression Analysis
Correlation Analysis: Rank Method & Karl Pearson's Coefficient of Correlation and Properties of Correlation.
Regression Analysis: Fitting of a Regression Line and Interpretation of Results, Properties of Regression Coefficients and Relationship between Regression and Correlation.

Unit IV (8 Sessions): Probability Theory & Distribution
Probability: Theory of Probability, Addition and Multiplication Law, Baye's Theorem
Probability Theoretical Distributions: Concept and application of Binomial; Poisson and Normal distributions. Introduction to bivariate and multivariate data analysis (Cluster and Factor analysis)

Unit V (8 Sessions) *Hypothesis Testing& Business Analytics*
Hypothesis Testing: Null and Alternative Hypotheses; Type I and Type II errors; Testing of Hypothesis: Large Sample Tests, Small Sample test, (t, F, Z Test and Chi Square Test)
Concept of Business Analytics- Meaning types and application of Business Analytics, Use of Spread Sheet to analyze data-Descriptive analytics and Predictive analytics.

Course Outcome:

Course Outcome	Blooms Taxonomy
CO1. Gaining Knowledge of basic concept / Fundamentals of business statistics.	• Knowledge (K 2)
CO2. To compute various measures of central tendency, Measures of Dispersion, Time Series Analysis, Index Number, Correlation and Regression analysis and their implication on Business performance.	• Remembering (K1) • Applying (K 4)
CO3. Evaluating basic concepts of probability and perform probability theoretical distributions	• Comprehending (K 3) • Applying (K 4)
CO4. To apply Hypothesis Testing concepts and able to apply inferential statistics- t, F, Z Test and Chi Square Test	• Analyzing (K 5) • Synthesizing (K6)
CO5. To perform practical application by taking managerial decision and evaluating the Concept of Business Analytics.	• Evaluating (K7) • Applying (K 4)

Suggested Readings

1. G C Beri – Business Statistics, 3rd ed, TATA McGrawHill.
2. Chandrasekaran & Umaparvathi-Statistics for Managers, 1st edition, PHI Learning
3. Davis, Pecar – Business Statistics using Excel, Oxford
4. Ken Black – Business Statistics, 5th ed., Wiley India
5. Levin and Rubin – statistics for Management, 7th ed., Pearson
6. Lind, Marchal, Wathen – Statistical techniques in business and economics, 13th ed, McGrawHill
7. Newbold, Carlson, Thorne – Statistics for Business and Economics, 6th ed., Pearson
8. S. C. Gupta – Fundamentals of Statistics, Himalaya Publishing
9. Walpole – Probability and Statistics for Scientists and Engineers, 8th ed., Pearson

MARKETING MANAGEMENT
BMBA105

Course Credit: 3 **Contact Hours: 40 hours**

Course Objectives

1. Assess market opportunities by analyzing customers, competitors, collaborators, context, and the strengths and weaknesses of a company.
2. Understand consumers' requirements and their behaviors.
3. Develop effective marketing strategies to achieve organizational objectives.
4. Communicate and defend your recommendations and critically examine and build upon the recommendations of your classmates both quantitatively and qualitatively.
5. Develop the understanding the current global and digital aspect of marketing.

Unit 1(6 hours)

Introduction: Nature and scope of marketing, Various marketing orientations, Need, Want, Demand, Elements of Marketing mix, customer value and the value delivery process.

Understanding Consumer Behavior: Buying motives, factors influencing buying behavior, buying habits, stages in consumer buying decision process, types of consumer buying decisions.

Unit 2 (8 hours)

Market segmentation, Targeting and Positioning: Meaning, Factors influencing segmentation, Market Aggregation, Basis for segmentation, Segmentation of Consumer. Targeting: Meaning, Basis for identifying target customers, Target Market Strategies. Positioning: Meaning, product differentiation strategies, tasks involved in positioning. Branding: Concept of Branding, Brand Types, Brand equity, Branding Positioning.

Unit 3 (8 hours)

Product Decisions: Concept, Product Hierarchy, New Product Development, Diffusion Process, Product Life cycle, Product mix strategies. Packaging / Labeling: Packaging as a marketing tool, requirement of good packaging, Role of labeling in packaging. Pricing Decisions: Pricing concepts for establishing value, Pricing Strategies-Value based, Cost based, Market based, Competitor based, New product pricing – Price Skimming & Penetration pricing

Unit 4 (8 hours)

Place Decision: Meaning, Purpose, Channel alternatives, Factors affecting channel choice, Channel design and Channel management decisions, Channel conflict, Retailing & Types of Retailers. Advertising: Advertising Objectives, Advertising Budget, Advertising Copy, AIDA model, Public Relation: Meaning, Objectives, Types, and Functions of Public Relations. Sales Promotion: Sales Promotion Mix, Kinds of promotion, Tools and Techniques of sales promotion, Push-pull strategies of promotion, Personal Selling: Concept, Features, Functions, Steps/process involved in Personal Selling, Direct Marketing: Meaning, Features, Functions, Growth and benefits of direct marketing, different forms.

Unit 5 (6 hours)

CRM: Meaning, Relationship Marketing Vs. Relationship Management, Types of Relationship Management, Significance of Customer Relationship Management. Global Marketing: current scenario, Global Marketing environment, Entry strategies, Global P's of Marketing., Recent trends and Innovation in Marketing- Green Marketing, Agile Marketing

Course Outcomes:

Upon the successful completion of this course, the student will be able to:

S.No	Course Outcome	Bloom's taxonomy
1	CO1. Remember and Comprehend basic marketing concepts.	• Remembering (K1) • Knowledge (K 2)
2	CO2. Understand marketing Insights on application of basic marketing concepts.	• Synthesizing (K6) • Comprehending(K3)
3	CO3. Able to Apply and develop Marketing Strategies and Plans	• Applying (K 4)
4	CO4. Understand and Analyzing Business/ Consumer Markets and ability Identify & evaluate Market Segments and Targeting	• Analyzing (K 5)
5	CO5. Develop skills to understand the current global and digital aspect of marketing.	• Evaluating (K7)

Recommended Text Books:

1. Marketing Management: A South Asian Perspective - Kotler, Keller, Kevin 15/e, Pearson Education, 2016.
2. Marketing Management - Ramaswamy V. S. & Namakumari S, 6/e, Sage Publication India Pvt Ltd., 2018.
3. Marketing Management - Tapan Panda, 5/e, Excel Publication, 2007.
4. Fundamentals of Marketing Management - Etzel M. J, B J Walker & William J. Stanton, 14/e, McGrawHill Education Publishers, 2015.
5. Marketing: Asian EditionPaul Bainies, Chris Fill Kelly Page third edition, Oxford.

INTRODUCTION TO BUSINESS ANALYTICS & DATA SCIENCE
BMBA106

Course Credit: 3 **Contact Hours: 40 hours**

Course Objectives:

1. Understanding the Role of Business Analyst and Data Science in business.
2. Understanding the basic concept of data management and data mining techniques
3. To understand the basic concept of machine learning
4. To understand the application of business analysis.
5. Understanding the basic concept of Data Science Project Life Cycle.

UNIT I (8 hrs.)
Introduction: What is business analytics? Historical Overview of data analysis, Data Scientist vs. Data Engineer vs. Business Analyst, Career in Business Analytics, what is data science, Why Data Science, Applications for data science, Data Scientists Roles and Responsibility

UNIT II (8 hrs.)
Data: Data Collection, Data Management, Big Data Management, Organization/sources of data, Types of Data, Importance of data quality, Dealing with missing or incomplete data, Data Visualization, Data Classification
Data Science Project Life Cycle: Business Requirement, Data Acquisition, Data Preparation, Hypothesis and Modeling, Evaluation and Interpretation, Deployment, Operations, Optimization.

UNIT III (8 hrs.)
Introduction to Data Mining, The origins of Data Mining, Data Mining Tasks, OLAP and Multidimensional data analysis, Basic concept of Association Analysis and Cluster Analysis.

UNIT IV (8 hrs.)
Introduction to Machine Learning: History and Evolution, AI Evolution, Statistics V/S Data Mining V/S Data Analytics V/S, Data Science, Supervised Learning, Unsupervised Learning, Reinforcement Learning, Frameworks for building Machine Learning Systems.

UNIT V (8hrs.)
Application of Business Analysis: Retail Analytics, Marketing Analytics, Financial Analytics, Healthcare Analytics, Supply Chain Analytics.

Course Outcomes: Upon the successful completion of this course, the student will be able to:

Course Outcome	**Bloom's Taxonomy**
CO1. Understand the basics of business analysis and Data Science	Knowledge (K2)
CO2. Understand data management and handling and Data Science Project Life Cycle	Comprehending (K3)
CO3. Understand the data mining concept and its Techniques	Applying (K4)
CO4. Understand and Analyzing machine learning concept	Analyzing (K5)
CO5. Understand the application of business analysis in different domain	Applying (K4)

Suggested Readings

1. Essentials of Business Analytics: An Introduction to the methodology and its application, Bhimasankaram Pochiraju, Sridhar Seshadri, Springer
2. Introduction to Machine Learning with Python: A Guide for Data Scientists 1st Edition, by Andreas C. Müller, Sarah Guido, O'Reilly
3. Introduction to Data Science, Laura Igual Santi Seguí, Springer
4. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Vipin Kumar, Pearson Education India
5. An Introduction to Business Analytics, Ger Koole, Lulu.com, 2019

IT SKILLS-1
BMBA151

Course Credit: 3 **Contact Hours: 40**

Course Objectives

1. To provide knowledge about the functioning of computers and its uses for managers
2. To provide hands on learning on Internet and its applications
3. To provide hands on learning on Word processing software
4. To provide hands on learning of applications on Spreadsheet software
5. To provide hands on learning on Presentation software

UNIT I: (05 hours) Conceptual Framework

Hardware: (a) Input devices - keyboard, printing devices, voice speech devices, scanner, MICR, OMR, Bar code reader, digital camera etc. (b) Output devices - Visual Display UNIT, printers, plotters (c) Storage Devices – Magnetic storage devices, Optical storage devices, Flash Memory.
Software: Types of software with examples; Introduction to languages, compiler, interpreter and Assembler, Operating System Functions, Types and Classification, Elements of GUI based operating system. Network and Internet: Types of computer networks (LAN, WAN and MAN), Netiquettes, Basic services over Internet like WWW, FTP, Telnet, Gopher, URL, Domain names, Web Browsers, Multimedia and its applications: Concepts of Text, Graphics, Animation, Audio, Images, Video. Multimedia Application in Education, Entertainment, Marketing. Names of common multimedia file formats,

UNIT II: Windows and Users Interface (Lab Work)- 7 hours

Windows operating System: Introduction and characteristics, Elements of GUI. Using Mouse, My Computer Icon, The Recycle Bin, Status Bar, Start and Menu & Menu-selection, Running an Application, Windows Explorer: Viewing of File, Folders and Directories Creating and Renaming of files and folders Opening and closing of different Windows, Windows Setting: Control Panels, Wall paper and Screen Savers Setting the date and Sound. Concept of menu, Using Help, Using right Button of the Mouse, Creating Short cuts, Basics of Window Setup, Notepad, Window Accessories

UNIT III: Word Processor Software (Lab Work) – 8 hours

Word processing concepts: Opening, Saving, Closing the file, opening an existing document, Selecting text, Editing text, Finding and replacing text, printing documents, Creating and Printing Merged Documents, Character and Paragraph Formatting, Page Design and Layout. Editing and Profiling Tools: Checking and correcting spellings. Using Graphics, Tables, Charts, Document Templates and Wizards.

UNIT IV: Spreadsheet Software (Lab Work) – 10 hours

Spreadsheet Package Spreadsheet: Concept and Working Interface, Creating, Saving and Editing a Workbook, Inserting, Deleting Work Sheets, entering data in a cell / formula Copying and Moving from selected cells, handling operators in Formulae. Functions in Spreadsheet: Mathematical, Logical, statistical, text, financial, Date and Time functions, Using Function Wizard. Formatting a Worksheet and Cell: changing data alignment, changing date, number, character or currency format, changing font, adding borders and colors. Printing worksheets, Charts and Graphs – Creating, Previewing, and Modifying Charts. Integrating word processor, spread sheets, web pages.

UNIT V: Presentation Software (lab Work) – 8 hours

Interface of the Presentation Package: Creating, Opening and Saving Presentations. Professional Look of the Presentation: Working in different Design & Views, Working with Slides. Formatting and Editing: Text, Image and Paragraph formatting, Checking Spelling and Correcting Typing Mistakes, Making

Notes Pages and Handouts, Drawing and Working with Objects, Adding Clip Art and other pictures, Designing Slide Shows, Running and Controlling a Slide Show, Printing Presentations.

Course Outcomes

Upon successful completion of this course, the student should be able to:

S. No.	Course Outcome	Bloom's Taxonomy
1	CO1. Gain in depth knowledge about the functioning of computers and its uses for managers	Knowledge (K2)
2	CO2. Learn to use Internet and its applications	Applying (K4)
3	CO3. Understand and implement Word processing software	Synthesizing (K6)
4	CO4. Learn applications on Spread sheet Softwares	Applying (K4) Knowledge (K2)
5	CO5. Analyse and learn Presentation software	Analyse (K5)

Suggested Readings

1. Nasib Singh Gill – Handbook of Computer Fundamentals, Khanna Publishing House, Delhi
2. Shrivastava-Fundamental of Computer& Information Systems (Wiley Dreamtech)
3. Leon A and Leon M - Introduction to Computers (Vikas, 1st Edition).
4. ITL ESL – Introduction to Information Technology (Pearson, 2nd Edition).
6. Introduction to Computers, Norton P. (TATA McGraw Hill)
7. Leon - Fundamentals of Information Technology, (Vikas)
8. Satish Jain-BPB's Computer Course Windows 10 with MS Office 2016 (BPB)
9. Linda Foulkes- Learn Microsoft Office 2019: A comprehensive guide to getting started with Word, PowerPoint, Excel, Access, and Outlook (Packt Publishing Limited)

Basics of Data Management with "R"
BMBA152

Course Credit: 3 **Contact Hours: 40 hours**

Course Objectives

1. Develop an understanding of the basic syntax and operations in R programming.
2. Equip students with the ability to organize, clean, and modify data using R's data frames and dplyr package.
3. Teach students how to create various data visualizations using the ggplot2 package in R.
4. Provide knowledge on applying descriptive statistics in R to summarize and describe data distributions.
5. Enable students to perform advanced data analysis and hypothesis testing using R.

UNIT I: Introduction to R and Data Handling (8 hrs.)
Learn the basics of R Syntax: Introduction to RStudio, R data types (vectors, lists, matrices, data frames, and factors), basic R commands, writing R scripts, and understanding the workspace. Loading and exploring datasets, sub-setting and filtering data, renaming and reordering columns, and creating new variables using the dplyr package.

UNIT II: Data Preparation in R (6 hrs.)
Learn how to prepare data for analysis in R using dplyr and tidyr: Data cleaning techniques, handling missing values, data transformation, reshaping data using pivoting (long to wide and vice versa), combining and splitting datasets, and data manipulation using the tidyr package (e.g., gather, spread, separate, unite).

UNIT III: Data Visualization in R (8 hrs.)
Learn the basics of how to create visualizations using the popular R package ggplot2**:** creating different types of plots (scatter plots, line plots, bar charts, histograms, and boxplots), customizing plots (titles, labels, colors, themes), adding layers (geoms), faceting, and saving and exporting plots. Explore advanced visualization techniques like creating multi-panel plots, and handling overplotting using transparency and jittering.

UNIT IV: Descriptive Statistics in R (10 hrs.)
Introduction to summarizing data, using functions like summarize(), group_by(), and mutate(). Understanding central tendency, calculating mean, median, mode, and interpreting results in the context of datasets. Understanding and calculating variance, standard deviation, and their importance in data analysis. Understanding quartiles, quantiles, and how they describe data spread, using R functions to compute these statistics, and interpreting boxplots.

Unit 5: Advanced Data Analysis and Hypothesis Testing in R (8 hrs.)
Introduction to joins in R, different types of joins (inner, left, right, full), merging datasets using dplyr functions (inner_join, left_join, etc.), and working with multiple data sources. Introduction to hypothesis testing, types of t-tests (one-sample, independent two-sample, paired sample), assumptions of t-tests, conducting t-tests in R, interpreting results, and understanding p-values and confidence intervals. Explore the concept of effect size and its calculation.

Course Outcome

Course Outcome	Bloom's Taxonomy
CO1. Students will be able to write and execute basic R scripts to manipulate data and perform simple calculations.	Understand (K2)
CO2. Students will be able to effectively clean, organize, and transform datasets using data frames and dplyr in R.	Apply (K3)
CO3: Students will be able to create and customize a variety of data visualizations, such as scatter plots, bar charts, and histograms using ggplot2.	Create (K5)
CO4: Students will be able to calculate and interpret key descriptive statistics (mean, median, mode, variance, standard deviation, quartiles) in R.	Analyze (K4)
CO5: Students will be able to conduct hypothesis tests (e.g., t-tests) using R and interpret the results to make informed decisions based on data.	Evaluate (K6)

Suggested Readings

1. "The R Book" Michael J. Crawley, Wiley

2. "R in Action" -Robert L. Kabacoff, Dream Tech Press

3. "R for Data Science" – Hadley Wickham and Garrett Gorlemund, O'Reilly

4. "A First Course in Statistical Programming with R" - by W. John Braun Duncan J. Murdoch, Cambridge University Press

5. "Exploratory Data Analysis Using R" - by Ronald K. Pearson, CRC Press

6. "Beginning Data Science in R: Data Analysis, Visualization, and Modelling for the Data Scientist" - by Thomas Mailund, Apress

BUSINESS INTELLIGENCE USING POWER BI
BMBA153

Course Credit: 2 **Contact Hours: 30 hours**

Course Objectives

1. To introduce students to the fundamentals of Business Intelligence (BI) and Power BI, enabling them to understand the architecture and interface of Power BI for effective data visualization and analysis.
2. To equip students with the skills to connect, load, and transform data using Power Query Editor, facilitating the management and preparation of data for analysis in Power BI.
3. To develop students' understanding of data modeling concepts and techniques, including the creation of relationships between data tables, to build a robust data model in Power BI.
4. To enable students to create and utilize DAX expressions for calculating and analyzing data within Power BI, enhancing their ability to perform complex data analysis.
5. To teach students how to effectively visualize data using Power BI's various tools and features, culminating in the ability to publish reports and create interactive dashboards.

UNIT I: Introduction to Power BI (3 hrs.)
What is BI (Business Intelligence), What is Power BI and Why choose Power BI? Power BI VS Power Excel, Installing Power BI Desktop, Power BI Interface, Report View Tab, Data View Tab, Power BI Architecture.

UNIT II: Connecting and Shaping Data (5 hrs.)
Data Load in Power BI, why to use Power BI Query Editor? How to open or access Power Query Editor, Power Query Editor Interface, Manage Rows and columns, Text, Date and Number Transformations, grouping by using Power Query Editor, Pivot & Unpivot using Power Query Editor, Merge and Append Query using Power Query Editor.

UNIT III: Data Model in Power BI (6 hrs.)
What is Data Modeling? Database Normalization, Primary Key vs Foreign Key, Active and Inactive Relationship, Data Tables and Fact tables, Star Schema, Direction of Relationship, Relationship Cardinality, Connecting Multiples Tables

UNIT IV: Creating Measures with Dax (6 hrs.)
What is Dax (Data Analysis Expression)? Purpose and usage of DAX, Calculated Columns, Measures, Difference Between Calculated Column and Measures, Calculate Function, Implicit Measures and Explicit Measures, Time Intelligence Dax Function, Filter Dax Functions, Table Manipulation Dax Functions, Iterate Functions, Logical Dax Functions, Relationship Functions

UNIT V: Data Visualization in Power BI (10 hrs.)
What is Data Visualization? Data Visualization Using Power BI, Matrix Visual in Power BI, Conditional Formatting, Charts, How to Publish reports and creation Work Space? Dashboard Creation and Setting Alerts using a case study.

Course Outcome

Course Outcome	Bloom's Taxonomy
CO1: Students will be able to explain the concept of Business Intelligence (BI) and navigate the Power BI interface effectively.	Understand (K2)
CO2: Students will be able to perform data connections, load data, and apply transformations using Power Query Editor.	Apply (K3)
CO3: Students will be able to design and implement data models, including creating and managing relationships in Power BI.	Analyze (K4)
CO4: Students will be able to create and apply DAX formulas for advanced data calculations and analysis in Power BI.	Create (K5)
CO5: Students will be able to design and publish interactive dashboards and reports, applying data visualization techniques.	Evaluate (K6)

Suggested Readings

1. Introducing Microsoft Power BI -by Alberto Ferrari (Author), Marco Russo (Author), Microsoft Press;

2. Learn Power BI: A Comprehensive, Step-by-step Guide for Beginners to Learn Real-world Business Intelligence, By Greg Deckler, Packt Publishing

3. Microsoft Power BI For Dummies -by Jack A. Hyman, Wiley

4. Microsoft Power BI Dashboards Step by Step by Errin O'Connor, Microsoft Press;

5. Mastering Microsoft Power BI: Expert techniques for effective data analytics and business intelligence -by Brett Powell, Packt Publishing

6. Learning Microsoft Power Bi: Transforming Data into Insights by Jeremey Arnold, O'Reilly Media

BVA 151 : SPORTS AND YOGA

Objective of the Course:

- To maintain mental and physical wellness upright and develop ability in the students to cope up with the stress arising in the life.
- To create space in the curriculum to nurture the potential of the students in sports/games/yoga etc.

To introduce a practice oriented introductory course on the subject. More involved / advanced course may come up in subsequent years of study

PART A : SPORTS & GAME

Some form of Athletics would be compulsory for all students, unless restricted due to medical / physical reasons. In addition to this, student has to opt for at least one game out of the remaining mentioned below.

A fair theoretical knowledge and a reasonable amount of field / site practice of the chosen games will be essential.

1. Athletics **Compulsory**
2. Volleyball
3. Basketball
4. Handball
5. Football
6. Badminton
7. Kabaddi
8. Kho-kho
9. Table tennis
10. Cricket

PART B: YOGA

a. Introduction of Yoga

Introduction of Yoga, Origin of Yoga, Aims and Objective of Yoga, Patanjali Yoga darshan, Hath yoga, Gheranda Samhita, Karm yoga, Gyan yoga.

b. Asanas, Pranayam and Meditation Practices

Meaning of Asanas, Objective of Asanas, rules and regulations of Asanas and Pranayams, Types of Yogasana

Yogic postures: Standing Posture, Sitting posture, Supine posture, Prone posture, balancing Postures, Pranayam according to Patanjali and Hath Yoga, Meditaion Mudras.

c. Science of *effects* Yoga

Physiological of Asanas- Paranayama and meditation, stress management and yoga, Mental health and yoga practice, Health and Personality Development.

General Guidelines

1. Institutes must assign minimum of three periods in the Time Table for the activities of Sports/Yoga.
2. Institutes must provide field/facility and offer a minimum of five choices of the Games/Sports.
3. Institutes are required to provide sports instructor / yoga teacher to mentor the students.

Student must be made familiar with the terminologies, rules/regulations, dimension/ marking of the play field/area and general knowledge of national/ international level facts/figures related to the chosen game.

SEMESTER II

EXPLORATORY AND PREDICTIVE ANALYTICS USING PYTHON
BMBA201

Course Credit: 3 **Contact Hours: 40 hours**

Course Objectives

1. To introduce students to the basics of Python programming, focusing on fundamental concepts and building a foundation for data analysis.
2. To equip students with the skills to effectively handle and manipulate data using Pandas, essential for data preparation and analysis.
3. To develop students' ability to perform exploratory data analysis, focusing on summarizing, visualizing, and interpreting data to gain insights.
4. To introduce students to predictive analytics and basic machine learning concepts, providing an understanding of how to build and evaluate predictive models.
5. To provide students with hands-on experience in applying exploratory and predictive analytics techniques to real-world business problems.

UNIT I (8 hrs.)

Introduction to Python: Overview of Python, its applications in business analytics, and setting up Python environment (Anaconda, Jupyter Notebook). Basic Python Syntax: Variables, data types, operators, and expressions. Control Structures: Conditional statements (if, else, elif), loops (for, while), and iteration techniques. Functions and Modules: Defining functions, using built-in functions, importing and using Python libraries. Basic Input/Output Operations: Reading from and writing to files, handling user input.

UNIT II (8 hrs.)

Introduction to Pandas: Understanding data structures in Pandas (Series, Data Frames). Data Importing and Exporting: Reading data from CSV, Excel, and other file formats, writing data to files. Data Cleaning: Handling missing data, removing duplicates, and data filtering. Data Manipulation: Sorting, indexing, and grouping data, merging and concatenating Data Frames. Working with Dates and Times: Handling date and time data, extracting and modifying date and time components.

UNIT III (8 hrs.)

Introduction to EDA: Importance of EDA in business analytics, steps in performing EDA. Descriptive Statistics: Calculating and interpreting mean, median, mode, variance, standard deviation, and quartiles. Data Visualization with Matplotlib and Seaborn: Creating basic plots (line, bar, histogram, boxplot), customizing plots (titles, labels, colors). Advanced Visualization Techniques: Pair plots, heatmaps, and correlation matrices. Identifying Patterns and Trends: Analyzing relationships between variables, detecting outliers, and understanding distributions.

UNIT IV (8 hrs.)

Introduction to Predictive Analytics: Understanding predictive analytics, use cases in business. Introduction to Machine Learning: Overview of supervised vs. unsupervised learning, key concepts (features, labels, training, testing). Linear Regression: Understanding and implementing simple linear regression, evaluating model performance (R-squared, MAE, RMSE). Introduction to Classification: Basics of classification, implementing logistic regression, confusion matrix, accuracy, precision, recall. Introduction to Scikit-Learn: Using Scikit-Learn for building and evaluating machine learning models.

UNIT V (8 hrs.)

Case Study 1: Sales Data Analysis: Performing EDA and visualization on sales data, identifying key trends and patterns. Case Study 2: Customer Segmentation: Using clustering techniques (e.g., K-Means) for customer segmentation and analysis. Case Study 3: Predictive Modeling: Building a predictive model for customer churn, interpreting results, and making business recommendations.

Course Outcome

Course Outcome	Bloom's Taxonomy
CO1: Students will be able to write simple Python programs using basic syntax, control structures, and functions.	Apply (K3)
CO2: Students will be able to clean, manipulate, and transform data in Pandas DataFrames, preparing datasets for analysis.	Analyze (K4)
CO3: Students will be able to perform exploratory data analysis, including calculating descriptive statistics and creating visualizations to identify patterns in data.	Evaluate (K6)
CO4: Students will be able to build and evaluate a simple predictive model using linear regression, understanding key metrics like R-squared and RMSE.	Create (K5)
CO5: Students will be able to apply analytics techniques to a business case study, interpret the results, and make data-driven recommendations.	Apply (K3)

Suggested Readings:

1. Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython" by Wes McKinney, O'Reilly Publication
2. Introduction to Machine Learning with Python: A Guide for Data Scientists" by Andreas C. Müller and Sarah Guido, O'Reilly Pulication
3. Python Data Science Handbook: Essential Tools for Working with Data by Jake Vander Plas, Shroff/O'Reilly Publication
4. Python Crash Course: A Hands-On, Project-Based Introduction to Programming" by Eric Matthes
5. Hands-On Data Analysis with Pandas by Stefanie Molin
6. Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking by Foster Provost and Tom Fawcett, Kindle Edition
7. Think Stats: Exploratory Data Analysis by Allen B. Downey, O'Reilly Publication
8. An Introduction to Statistical Learning with Applications in R" by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, Springer
9. Practical Statistics for Data Scientists: 50 Essential Concepts by Peter Bruce and Andrew Bruce, O'Reilly Publication
10. Machine Learning using Python by Manaranjan Pradhan and U. Dinesh Kumar, Wiley

D**ATA MINING TECHNIQUES – PREDICTIVE MODELING & PATTERN DISCOVERY- USING R**
BMBA202

Course Credit: 3 Contact Hours: 40 Hrs.

Course Objectives:

- Understanding of data mining and its functions
- Understanding of classification, clustering algorithms
- To apply classification and clustering methods applicable to predictive analytics using R
- Understanding of how to formulate predictive analytics using R
- Understand pattern discovery using R

UNIT I (8 hrs.)
Data Mining: Overview, Motivation, Definition & Functionalities, Data Processing, Form of Data Preprocessing, Data Cleaning.: Missing Values, Noisy Data, (Binning, Clustering, Regression, Computer and Human inspection), Inconsistent Data, Data Integration and Transformation. Data Reduction: Data Cube Aggregation, Dimensionality reduction, Data Compression.

UNIT II (8 hrs.)
Classification: Definition, Data Generalization, Analytical Characterization, Analysis of attribute relevance, Mining Class comparisons, Statistical measures in large Databases, Statistical-Based Algorithms, Distance-Based Algorithms, Decision Tree-Based Algorithms. Clustering: Introduction, Similarity and Distance Measures, Hierarchical and Partitional Algorithms. Hierarchical Clustering-CURE and Chameleon. Association rules: Introduction, Large Item sets, Basic Algorithms, Parallel and Distributed Algorithms, Neural Network approach

UNIT III (8 hrs.)
Data Mining process- CRISP -DM Methodology, Data Collection and Business understanding, Data and Datasets, importing data into R, Data Preprocessing: Data Cleaning, Transforming variables, creating variables, Dimensionality Reduction, Modeling: Exploratory data analysis, dependency modeling using association rules, clustering, anomaly detection,

UNIT IV (8 hrs.)
Predictive analytics-Evaluation Metrics, Tree-Based Model, Support Vector Machines, Artificial Neural Networks and deep learning, Model Ensembles, Evaluation- The holdout and random subsampling, cross validation, bootstrap estimates, recommended procedures, reporting and deployment, Case Study.

UNIT V (8 hrs.)
Transactional Dataset, Apriori Analysis, Generating Filtering Rules, Plotting, Sequential Dataset, Apriori Sequence Analysis, Understanding the Results, Business Cases

Course Outcome: After successful completion of this course students will be able to-

Course Outcomes	Bloom's Taxonomy
CO1: Understanding of data mining and its functions	Knowledge (K2) Remembering(K1)
CO2: Understanding of classification, clustering algorithms	Comprehending(K3) Analyzing(K5)
CO3: To apply classification and clustering methods applicable to predictive analytics using R	Applying(K4)
CO4: Understanding of how to formulate predictive analytics using R	Applying(K4)
CO5: Understand pattern discovery using R	Analyzing (K5) Applying(K4)

Suggested Readings:

1. Data Mining with R: Learning with Case Studies, Luís Torgo, Chapman and Hall/CRC; 2nd edition
2. An Introduction to Statistical Learning with Applications in R" by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, Springer
3. R Data Mining: Implement data mining techniques through practical use cases and real-world datasets, Andrea Cirillo, Packt Publishing; 1 edition
4. R Data Science Essentials, By Raja B. Koushik, Sharan Kumar Ravindran, Packt Publishing

BUSINESS RESEARCH METHODS
BMBA 203

Course Credit: 3 **Contact Hours:40**

Course objectives

1. Understand the concept / fundamentals of research and their types.
2. Understand the practical application of various research techniques.
3. Understand the importance of scaling & measurement techniques and sampling techniques
4. Understand the importance of coding, editing, tabulation and analysis in doing research.
5. Understanding and applying the concept of statistical analysis which includes ANOVA technique and technique of report writing.

Unit 1 (8 Sessions)

Research: – Definition, Meaning, Importance types and Qualities of Research; Research applications in functional areas of Business, Emerging trends in Business research.

Research & the Scientific Method: Characteristics of scientific method. Steps in Research Process

Concept of Scientific Enquiry: – Formulation of Research Problem – Management Question – research Question – Investigation Question

Research Proposal – Elements of a Research Proposal, Drafting a Research Proposal, evaluating a research proposal.

Unit 2 (8 Sessions)

Research design: Concept, Features of a good research design, Use of a good research design; Qualitative and Quantitative research approaches, Comparison – Pros and Cons of both approaches.

Exploratory Research Design: Concept, Types: Qualitative techniques – Projective Techniques, Depth Interview, Experience Survey, Focus Groups, Observation.

Descriptive Research Designs: Concept, types and uses. Concept of Cross-sectional and Longitudinal Research

Experimental Design: Concept of Cause, Causal relationships, Concept of Independent & Dependent variables, concomitant variable, extraneous variable, Treatment, Control group.

Unit 3 (6 Sessions)

Scaling & measurement techniques: Concept of Measurement: Need of Measurement; Problems in measurement in management research – Validity and Reliability. Levels of measurement – Nominal, Ordinal, Interval, Ratio. Attitude Scaling Techniques: Concept of Scale – Rating Scales viz. Likert Scales, Semantic Differential Scales, Constant Sum Scales, Graphic Rating Scales – Ranking Scales – Paired comparison & Forced Ranking – Concept and Application.

Unit 4 (6 Sessions)

Sampling: Basic Concepts: Defining the Universe, Concepts of Statistical Population, Sample, Characteristics of a good sample. Sampling Frame (practical approach for determining the sample frame expected), Sampling errors, non-sampling errors, Methods to reduce the errors, Sample Size constraints, non-response.

Probability Sample: Simple Random Sample, Systematic Sample, Stratified Random Sample, Area Sampling & Cluster Sampling.

Non-Probability Sample: Judgment Sampling, Convenience Sampling, Purposive Sampling, Quota Sampling & Snowballing Sampling methods. Determining size of the sample – Practical considerations in sampling and sample size, sample size determination.

Unit 5 (8 Sessions)

Data Analysis: Editing, Coding, Tabular representation of data, frequency tables, Construction of frequency distributions, Graphical Representation of Data: Appropriate Usage of Bar charts, Pie charts, Histogram.

Hypothesis: Qualities of a good Hypothesis –Framing Null Hypothesis & Alternative Hypothesis. Concept of Hypothesis Testing – Logic & Importance. Analysis of Variance: One way and two-way Classifications.

Mechanism of Report Writing- Report Preparation: Types, Report Structure: preliminary section, main report, interpretation of results, suggestions and recommendations, limitations of the study, Report formulation.

COURSE OUTCOME

Course Outcomes	Blooms Taxanomy
CO1. Knowledge of concept / fundamentals for different types of research.	• Knowledge (K 2)
CO2. Applying relevant research techniques.	• Remembering (K1) • Applying (K 4)
CO3. Understanding relevant scaling & measurement techniques and should use appropriate sampling Techniques	• Comprehending (K 3) • Applying (K 4)
CO4.Synthesizing different techniques of coding, editing, tabulation and analysis in doing research.	• Analyzing (K 5) • Synthesizing (K6)
CO5.Evaluating statistical analysis which includes ANOVA technique and prepare research report.	• Evaluating (K7)

Suggested Readings

1. Research Methodology, Deepak Chawla, Neena Sondhi, Vikas Publication
2. Business Research Methods, Naval Bajpai, Pearson Education
3. Research Methodology, C R Kothari, New Age International.
4. Business Research Methods by Donald Cooper & Pamela Schindler, TMGH, 9th Edition.
5. Business Research Methods by Alan Bryman & Emma Bell, Oxford University Press, 2ndEdition.
6. Business Research Methods by T N Srivastava & Shailaja Rao, TMH Publication, 2ndEdition.

FINANCIAL MANAGEMENT AND CORPORATE FINANCE
BMBA 204

Course Credit: 3 **Contact Hours: 40 Hrs.**

Course Objectives: This course is intended to introduce the basic theory, concepts and practical applications in corporate finance and to enable students to analyse various corporate decisions. The course objectives are outlined below:

1) To understand the fundamentals, various models and agency problems of Corporate Finance.
2) To acquire knowledge about various techniques used for analysing various long-term projects.
3) To have an understanding about various capital structure techniques and selecting best source of finance.
4) To have an understanding of various dividend models and its applicability.
5) To acquaint students about corporate valuation in mergers and acquisitions.

UNIT I (8 Hrs)

Introduction to Finance & Corporate Finance: Corporate Finance & its scope, Corporate Governance and Agency Problem, Finance & Corporate Strategy, Time Value of Money, Risk and Return. Types of Financial Markets: Capital Market, Factors affecting Financial Markets, Linkages between Economy & Financial Markets, Integration of Indian Financial Markets with Global Financial Markets.

UNIT II (8 Hrs)

Investment and Financing Decision: Capital Budgeting, Nature of investment decisions, Risk Analysis in investment decisions, Concept of Opportunity Cost, Cost of Debenture, Preference and Equity capital, Composite Cost of Capital, Cash Flows as Profit and components of Cash Flows, Capital Budgeting Decisions, Calculation of NPV and IRR, Excel Application in Analysing Projects.

UNIT III (8 Hrs)

Financial Decision: Capital Structure, Relevance and Irrelevancy theory, Leverage analysis – financial, operating and combined leverage along with its implications, EBIT EPS Analysis, Point of Indifference.

UNIT IV (8 Hrs)

Dividend Relevance: Factors affecting Dividend Policy, Forms of Dividends, Types of Dividend Policies, Dividend Models: Walter and Gordon Model, Miller- Modigliani (MM) Hypothesis. Theories of Dividend Payout (Dividend Irreverence Theory, Bird- in the hand Theory, Tax Preference Theory.

UNIT V (8 Hrs)

Working Capital Management : Concepts & Principles of Working Capital, Need for working capital, Classification and importance of working capital, Working capital cycle, Inventory Management, Cash Management, Accounts receivable Management and Factoring, Credit policy, Financing working capital.

Course Outcome: After successful completion of this course students will be able:

S.No.	Course Outcome	Bloom's Taxonomy
1.	CO1 Understand the different basic concept / Models of Corporate Finance and Governance	• Knowledge (K2) • Remembering(K1)
2.	CO2 Understand the practical application of time value of money and evaluating long term investment decisions	• Analyzing (K5) • Evaluating(K7)
3.	CO3 Develop analytical skills to select the best source of capital, structure and leverage.	• Analyzing(K5) • Synthesizing(K6)
4.	CO4 Understand the use and application of different models for firm's optimum dividend pay-out.	• Comprehending(K3) • Applying(K4)
5.	CO5 Understand the recent trends of mergers and acquisition and its valuation	• Comprehending(K3) • Synthesizing (K6)

Suggested Readings

1) Khan and Jain - Financial Management (Tata McGraw Hill, 7th Ed.)
2) Pandey I M - Financial Management (Vikas, 11th Ed.)
3) William HakkaBettnerCarcello- Financial and Management Accounting (TMH-16th Ed.)
4) Sheebakapil-Fundamental of financial management (Wiley,2015)
5) Prasanna Chandra - Fundamentals of Financial Management (TMH, 9th Ed.)
6) Bark DemazoThampy- Financial Management (Pearson,2nd Ed.)
7) R P Rustagi - Financial Management (Galgotia, 2000, 2nd revised ed.)
8) Damodaran, A., Applied Corporate Finance, 3rd Edition, Wiley, 2012
9) Ravi.M Kishore – Financial Management (Taxman, 7th Ed)
10) Fundamentals to Financial Management, Brigham & Houston, 14/e, Cengage Learning
11) Van Horne - Financial Management and Policy (Prentice hall, 2003, 12th Ed.)

DATABASE MANAGEMENT SYSTEM AND SQL
BMBA 205

Course Credit: 3 **Contact Hours: 40**

Course Objectives:

1. To provide a comprehensive understanding of database management systems (DBMS) and their importance in business applications.
2. To equip students with the skills to design, implement, and manage databases using SQL.
3. To enable students to retrieve and manipulate data using SQL for business decision-making.
4. To develop an understanding of database normalization, relationships, and indexing.
5. To prepare students to use advanced SQL techniques for complex queries and database management.

UNIT I (8 hrs.)
Introduction to Databases: Definition, characteristics, and types of databases (hierarchical, network, relational). DBMS Overview: Definition, advantages, components, and architecture of DBMS (schema, instances, and database languages). Data Models: Overview of different data models (hierarchical, network, relational, object-oriented). Introduction to Relational Databases: Basic concepts of relational databases, tables, rows, columns, and keys. Database Applications: Examples of database applications in business (CRM, ERP, e-commerce).
UNIT II (8 hrs.)
Entity-Relationship (ER) Model: Entities, attributes, relationships, ER diagrams, and converting ER diagrams to relational schemas. Relational Database Design: Designing relational schemas, primary keys, foreign keys, and constraints. Normalization: Concept of normalization, functional dependencies, normal forms (1NF, 2NF, 3NF, BCNF), and their importance in database design. Designing Relational Databases: Practical steps in designing a relational database, identifying entities, relationships, and attributes.
UNIT III (8 hrs.)
Introduction to SQL: Overview of SQL, history, and importance in database management. Basic SQL Queries: SELECT statement, filtering data with WHERE clause, sorting data using ORDER BY, and using DISTINCT to eliminate duplicates. Working with Multiple Tables: JOIN operations (INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL JOIN) to retrieve data from multiple tables. Data Aggregation: Aggregate functions (COUNT, SUM, AVG, MIN, MAX), GROUP BY and HAVING clauses. Subqueries: Writing subqueries in SQL and understanding their use in complex queries.
UNIT IV (8 hrs.)
SQL Functions: String functions, date functions, and mathematical functions. Views in SQL: Creating and managing views, advantages of using views in database management. Transactions in SQL: Understanding transactions, ACID properties, COMMIT, ROLLBACK, and SAVEPOINT commands. Indexes: Introduction to indexing, types of indexes, and their role in query optimization. Stored Procedures and Triggers: Introduction to stored procedures and triggers, creating and managing them in SQL.
UNIT V (8 hrs.)
Database Security: Importance of database security, user roles, privileges, and access control. Backup and Recovery: Techniques for database backup, recovery, and ensuring data integrity. Database Performance Tuning: Basic concepts of performance tuning, query optimization, and database maintenance. Case Study 1: Designing and implementing a database for a small business application (e.g., inventory management). Case Study 2: Analyzing a business dataset using SQL queries to generate insights and reports.

Course Outcome: After successful completion of this course students will be able:

Course Outcome	Bloom's Taxonomy
CO1: Students will be able to explain the fundamental concepts of DBMS and describe their applications in business contexts.	Understand (K2)
CO2: Students will be able to design a normalized relational database schema based on business requirements.	Apply (K3)
CO3: Students will be able to write basic SQL queries to retrieve, filter, and manipulate data from a relational database.	Apply (K3)
CO4: Students will be able to use advanced SQL techniques, such as indexing and transactions, to optimize database performance.	Analyze (K4)
CO5: Students will be able to administer a database, ensure its security, and apply SQL knowledge to solve real-world business problems.	Evaluate (K6)

Suggested Readings

1) Database System Concepts by Abraham Silberschatz, Henry F. Korth, and S. Sudarshan, McGraw-Hill
2) SQL in 10 Minutes, Sams Teach Yourself by Ben Forta, Kindle Edition
3) Learning SQL: Generate, Manipulate, and Retrieve Data by Alan Beaulieu, O'Reilly Publication
4) Fundamentals of Database Systems by Ramez Elmasri and Shamkant B. Navathe, Pearson
5) SQL Performance Explained by Markus Winand
6) The Practical SQL Handbook: Using SQL Variants by Judith S. Bowman, Sandra L. Emerson, and Marcy Darnovsky, Addison-Wesley Professional

QUANTITATIVE TECHNIQUES FOR MANAGERS
BMBA 206

Course Credit: 3 **Contact Hour: 40**

Course Objectives

1. Understand the importance of the use of OR application in decision Making environment
2. To formulate LPP and Obtain Graphical Solutions & Acquire General idea of the Simplex method.
3. To understand and solve transportation & assignment models.
4. To know optimal sequence model and understand concepts of queuing theory.
5. To identify right time for replacement of equipment and understand project management techniques

Unit I (6 Sessions)-Operations Research & Decision-Making Environments
Operations Research: - Uses, Scope and Applications of Operation Research in managerial decision-making. Decision-*making environments:* - Decision-making under certainty, uncertainty and risk situations; Decision tree approach and its applications.

Unit II (6 Sessions)-Linear Programming Problem ~~& Transportation Problem~~
Linear programming: Mathematical formulations of LP Models for product-mix problems; graphical and simplex method of solving LP problems.

Unit III (10 Sessions)- Transportation Problem & Assignment model

Transportation problem: Various methods of finding Initial basic feasible solution-North West Corner Method, Least Cost Method & VAM Method and optimal solution-Stepping Stone & MODI Method, Maximization Transportation Problem

Assignment model: Hungarian Algorithm and its applications, Maximization Assignment Problem.

Unit IV (10 Sessions)-Sequencing & Queuing Theory
Sequencing Problem: Johnsons Algorithm for n Jobs and Two machines, n Jobs and Three Machines, Two jobs and m - Machines Problems.
Queuing Theory: Characteristics of M/M/I Queue model; Application of Poisson and Exponential distribution in estimating arrival rate and service rate; Applications of Queue model for better service to the customers.

Unit V (8 Sessions) Project Management
Project Management: Rules for drawing the network diagram, Applications of CPM and PERT techniques in Project planning and control; GANTT Chart

Course Outcomes

CO1	Be able to understand the characteristics of different types of decision-making environments and the appropriate decision-making approaches and tools to be used in each type.	Knowledge (K2)/ Remembering (K1)
CO2	To formulate linear programming problem and to find optimal solution by graphical simplex method.	Knowledge (K2)
CO3	Be able to build and solve Transportation Models and Assignment Models also to solve game theory problems by understanding pure and mix strategies.	Applying (K 4)
CO4	To assign optimal sequence of difference jobs on different machines and develop understanding of queuing theory concepts.	Applying (K 4)
CO5	To implement replacement of equipment at right time and able to implement project management concepts like CPM, PERT to reduce cost and time.	Synthesizing (K6)/ Evaluating (K7)

Suggested Readings

1. R. Panneerselvam - Operations Research (PHI, 2nd Edition)
2. Sharma J K - Operations Research (Pearson, 3rd Edition
3. Apte-Operation Research and Quantitative Techniques (Excel Books)
4. S Kalawathy-Operation Research (Vikas IVth Edition)
5. Natarajan- Operation Research (Pearson)
6. Singh & Kumar—Operation Research (UDH Publisher edition 2013)
7. Taha Hamdy - Operations Research - An Introduction (Prentice-Hall, 9th edition)
8. Vohra - Quantitative Techniques in Management (Tata McGraw-Hill, 2nd)
9. Kothari - Quantitative Techniques (Vikas 1996, 3rd Edition).

DATA VISUALIZATION USING TABLEAU
BMBA251

Course Credit: 3 **Contact Hours: 40 Hrs.**

Course Objective:

1. To introduce students to the fundamentals of data visualization and the importance of visual analytics in business decision-making.
2. To provide hands-on experience with Tableau, enabling students to create, customize, and interpret basic data visualizations.
3. To teach students how to design simple interactive dashboards in Tableau for effective data presentation.
4. To equip students with the skills to connect to and prepare data for visualization in Tableau.
5. To enable students to use basic Tableau features to create insightful visualizations for business applications.

UNIT I (8 hrs.)

Introduction to Data Visualization: Importance and principles of data visualization, overview of common chart types (bar charts, line charts, pie charts). Overview of Tableau: Introduction to Tableau, its features, and how it is used in business analytics. Tableau Interface: Navigating Tableau's workspace, understanding Tableau's key components (sheets, dashboards). Connecting to Data: Importing data into Tableau from Excel and CSV files, understanding data types and structures. Basic Visualizations: Creating simple charts (bar charts, line charts, pie charts) using Tableau's Show Me feature.

UNIT II (8 hrs.)

Connecting to Data Sources: Understanding data connections, importing data from multiple files. Basic Data Cleaning: Handling missing data, filtering data, and simple data transformations. Hierarchies and Groups: Creating basic hierarchies, groups, and sets to organize data. Sorting and Filtering Data: Sorting data in visualizations, applying basic filters to refine the data displayed.

UNIT III (8 hrs.)

Creating Common Visualizations: Building bar charts, line charts, pie charts, and scatter plots. Using Colors and Labels: Customizing visualizations with colors, labels, and tooltips for better data interpretation. Simple Calculated Fields: Introduction to creating calculated fields for basic data analysis (e.g., total sales, profit margin). Combining Visualizations: Creating combined charts (e.g., bar and line chart) and understanding when to use them.

UNIT IV (8 hrs.)

Introduction to Dashboards: Understanding the purpose of dashboards, best practices for designing beginner-friendly dashboards. Creating Dashboards: Building a basic interactive dashboard using multiple visualizations. Dashboard Interactivity: Adding simple filters and actions to make dashboards interactive. Exporting and Sharing Dashboards: Exporting dashboards as images or PDFs, and sharing them with stakeholders.

UNIT V (8 hrs.)

Case Study 1: Creating a sales report dashboard to visualize key metrics like sales and profits. Case Study 2: Analyzing customer data to create a basic customer segmentation dashboard. Review and Practice: Hands-on practice with real-world datasets, applying the concepts learned throughout the course.

Course Outcomes:

Course Outcome	Bloom's Taxonomy
CO1: Students will understand the basic principles of data visualization and be able to explain the role of visual analytics in business contexts.	Understand (K2)
CO2: Students will be able to create and customize basic charts and graphs in Tableau, and interpret the visualized data.	Apply (K3)
CO3: Students will design and build simple interactive dashboards in Tableau to communicate data insights effectively.	Create (K5)
CO4: Students will connect to and prepare data from various sources in Tableau, including basic data cleaning and transformation.	Analyze (K4)
CO5: Students will apply fundamental Tableau features to create straightforward visualizations and analyses.	Apply (K3)

Suggested Readings:

1. Tableau For Dummies by Molly Monsey and Paul Sochan
2. Learning Tableau: A Beginner's Guide by Joshua N. Milligan, Packt Pub Ltd
3. Tableau 9: The Official Guide by George Peck
4. The Big Book of Dashboards: Visualizing Your Data Using Real-World Business Scenarios by Steve Wexler, Jeffrey Shaffer, and Andy Cotgreave, Wiley Publication
5. Practical Tableau: 100 Tips, Tutorials, and Strategies from a Tableau Zen Master by Ryan Sleeper, O'Reilly Publication

ANALYTICS PROJECT USING R/PYTHON
BMBA252

Course Credit: 2 **Contact Hours: 30 Hrs.**

Course Objectives:

1. To provide students with practical experience in executing a complete analytics project using R/Python.
2. To develop students' skills in data collection, cleaning, and preparation using R/Python.
3. To enable students to apply statistical analysis and basic machine learning techniques to solve real-world problems.
4. To teach students how to visualize and communicate their findings effectively.
5. To guide students in documenting and presenting their project in a professional manner.

UNIT I (6 hrs.)
Understanding Analytics Projects: Overview of the analytics project lifecycle, types of analytics projects. Problem Definition: Defining the business problem or research question, setting objectives, and determining the scope of the project. Project Planning: Developing a project plan, including timeline, milestones, resources, and deliverables. Data Requirements: Identifying data needs, sources of data, and methods for data collection. Tools Setup: Introduction to RStudio (for R) and Jupyter Notebook (for Python), setting up the working environment.

UNIT II (6 hrs.)
Data Collection: Techniques for collecting data from various sources (web scraping, APIs, CSV/Excel files). Data Cleaning: Handling missing data, removing duplicates, correcting errors, standardizing data formats. Data Preparation: Transforming data, feature engineering, and preparing data for analysis. Exploratory Data Analysis (EDA): Conducting EDA to understand the data, identifying patterns, and initial insights.

UNIT III (6hrs.)
Statistical Analysis: Applying descriptive statistics, hypothesis testing, correlation, and regression analysis. Introduction to Machine Learning: Overview of basic machine learning concepts and techniques. Model Building: Building and evaluating simple models (e.g., linear regression, decision trees). Interpreting Results: Analyzing model outputs and deriving actionable insights.

UNIT IV (6hrs.)
Data Visualization Principles: Importance of visualization in analytics, principles of effective visual communication. Creating Visualizations: Using R/Python libraries (ggplot2, matplotlib, seaborn) to create basic charts and graphs. Storytelling with Data: Crafting a narrative around the data, using visualizations to support the story. Presentation Skills: Best practices for presenting data-driven insights to stakeholders.

UNIT V (6hrs.)
Project Execution: Applying the skills learned to execute the project, from data collection to final analysis. Documentation: Writing a concise project report, including problem statement, methodology, analysis, and conclusions. Final Presentation: Preparing and delivering a professional presentation of the project findings. Project Submission: Submitting the final report, code, and presentation materials.

Course Outcome: After the completion of the course the students will be able to:

Course Outcome	**Bloom's Taxonomy**
CO1: Students will be able to define a clear analytics problem and develop a project plan, including data collection and analysis strategies.	Create (K5)
CO2: Students will gain proficiency in using R/Python for data collection, cleaning, and preparation.	Apply (K3)
CO3: Students will apply appropriate statistical and basic machine learning techniques to analyze data and derive insights.	Apply (K3)
CO4: Students will create visualizations that effectively communicate their findings.	Evaluate (K6)
CO5: Students will document their project process, results, and conclusions, and present them professionally.	Evaluate (K6)

Suggested Readings:

1. R for Data Science by Hadley Wickham and Garrett Grolemund (for R users)
2. Python for Data Analysis by Wes McKinney (for Python users)
3. Practical Statistics for Data Scientists: 50 Essential Concepts by Peter Bruce and Andrew Bruce
4. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow by Aurélien Géron (for Python users)
5. Introduction to Statistical Learning by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani (for R and Python users)
6. Storytelling with Data: A Data Visualization Guide for Business Professionals by Cole Nussbaumer Knaflic

IT SKILLS-2

BMBA253

20 Hours

Course Objective

1. To develop pivot table and understand the validating & auditing techniques
2. To understand different charting techniques in MS Excel
3. To understand different formatting techniques in MS Excel

Unit I (Lab work on spreadsheet)

Pivot Table: Developing Pivot Table, analyzing data using goal seek and solver, Scenarios Create named scenarios. Show, edit, delete scenarios, Creating a scenario summary report. Validating and Auditing: Set, edit validation criteria for data entry in a cell range like: whole number, decimal, list, date, time, Trace precedent, dependent cells. Identify cells with missing dependents. Creating applications in Spreadsheet and Macros.

Unit II (Lab work on spreadsheet) 15 Hours

Creating and formatting Charts: Understanding chart types, column chart, bar chart, line chart, pie chart, XY Scatter chart, Area chart, surface chart, bubble chart. Create a combined chart like: column and line, column and area. Change the chart type for a defined data series, Add, delete a data series in a chart, Re-position chart title, legend, data labels. Change scale of value axis: minimum, maximum number to display, major interval. Change display units on value axis without changing data source: hundreds, thousands, millions. Format columns, bars, pie slices, plot area, chart area to display an image.

References

Excel Data Analysis: Modeling and Simulation, Hector Guerrero (Springer)

COURSE OUTCOME

S. No.	Course Outcome	Bloom's Taxonomy
1	CO1. To gain knowledge of pivot table and understand the validating & auditing techniques	Knowledge (K2)
2	CO2. Learn to use different charting techniques in MS Excel	Applying (K4) Synthesizing (K6)
3	CO3. Learn to use different formatting techniques in MS Excel	Applying (K4) Knowledge (K2)

MINI PROJECT
BMBA254

Course Credit: 2 **Contact Hours: 30 hrs**

Objective:

1. To identify the key issues and challenges within a selected industry.
2. To prepare a comprehensive report on the application of business analytics in the chosen industry.

Project Overview:

During the second semester, Business Analytics students are required to select an industry of their interest for an in-depth analysis and prepare an individual project report. The focus should be on identifying industry-specific challenges and exploring the role of **business analytics** in addressing these challenges.

Key Components of the Report:

1. **Industry Challenge Identification**: Analyze and identify the critical issues faced by the selected industry.
2. **Application of Emerging Technologies**: Investigate how emerging technologies are being, or can be, applied to solve these industry-specific challenges. Highlight case studies, use cases, or real-world applications where possible.
3. **Feasibility Analysis**:
    - **Market Feasibility**: Conduct a market analysis to understand industry trends, opportunities, and competitive landscape.
    - **Technical Feasibility**: Assess the practicality and readiness of the industry to adopt emerging technologies.
    - **Financial Feasibility**: Analyze the financial implications, including cost-benefit analysis and financial sustainability of the proposed solutions.

Evaluation:

The report will be prepared individually by each student and will be evaluated by an **external examiner** appointed by the university. The focus will be on the depth of analysis, application of business analytics concepts, and the feasibility of the proposed technological solutions.

COURSE OUTCOME

S. No.	Course Outcome	Bloom's Taxonomy
1	CO1. To gain knowledge of issues challenge of the industry	Knowledge (K2)
2	CO2. Learn to prepare report on the application of emerging technologies in the selected industry	Applying (K4) Synthesizing (K6)